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S2**Started on** Monday, 24 August 2026, 5:28 PM**State** Finished**Completed on** Monday, 24 August 2026, 5:37 PM**Time taken** 8 mins 56 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main() {
4      int num;
5      scanf("%d", &num);
6
7      int count = 0;
8
9      for (int i = 1; i <= num; ++i) {
10         count++; // i <= num condition check
11         count++; // if (num % i == 0) condition check
12
13         if (num % i == 0) {
14             count++; // printf("%d ", i);
15         }
16     }
17     count++; // final i <= num condition check (false)
18
19     printf("%d\n", count);
20
21     return 0;
22 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)